

Parallel Robots

EPFL – 2026

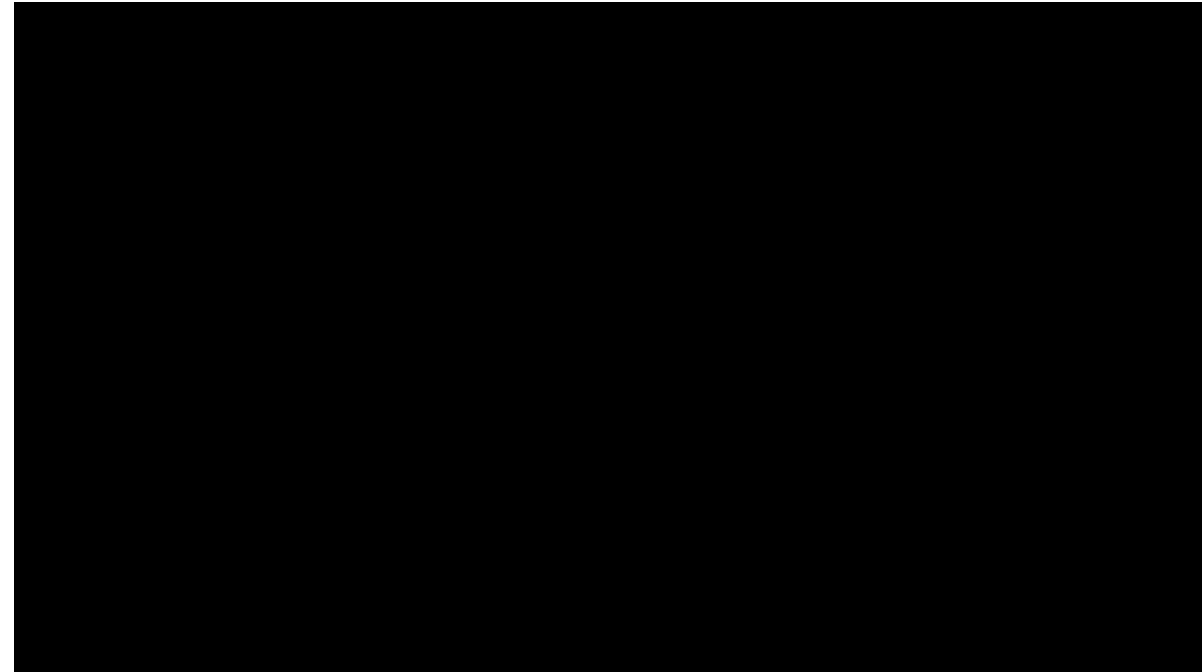
Arda YIĞIT



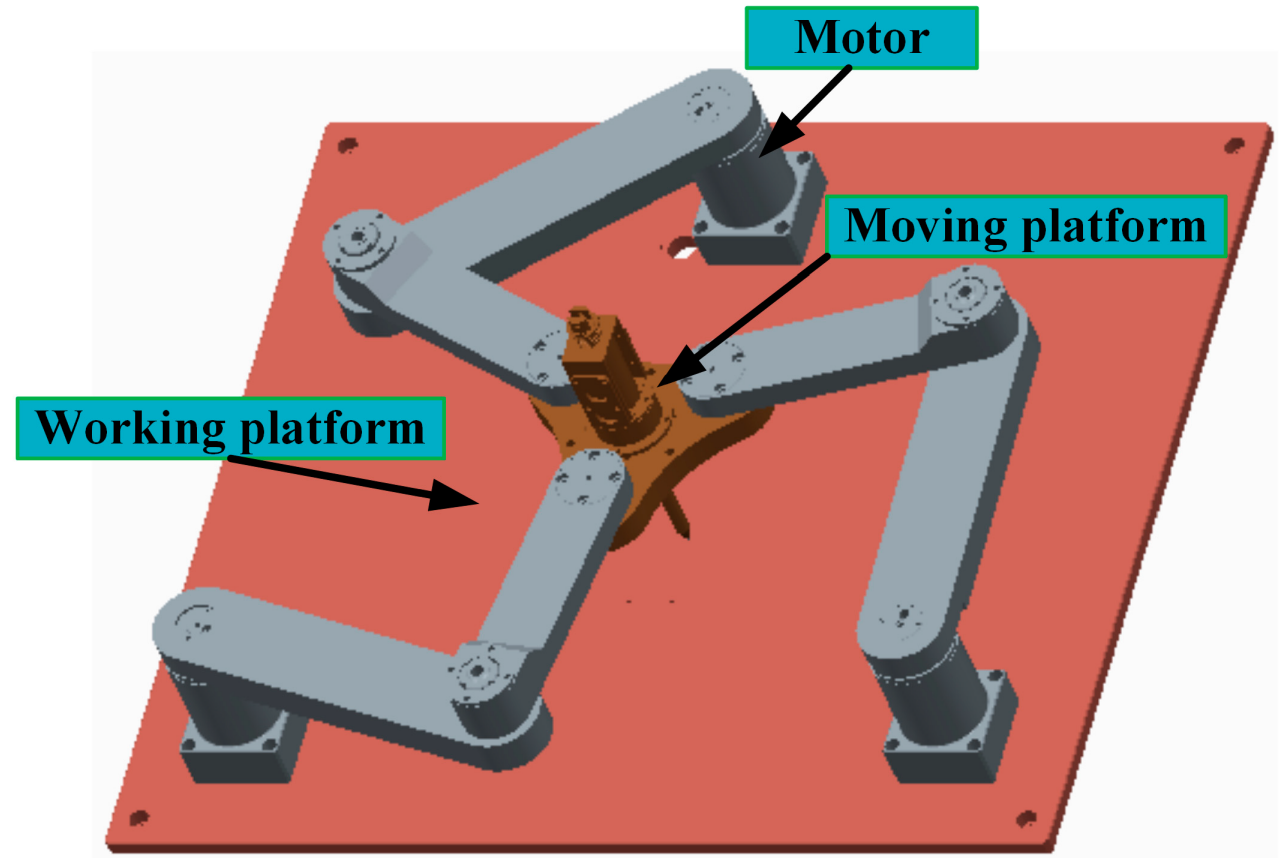
- Jean-Pierre Merlet's book
 - J.-P, Merlet, Parallel robots. Springer-Verlag, 2006. <https://doi.org/10.1007/1-4020-4133-0>
- Sébastien Briot's and Wisama Khalil's book
 - S. Briot and W. Khalil, Dynamics of Parallel Robots. Springer International Publishing, 2015. <https://doi.org/10.1007/978-3-319-19788-3>

Preliminary Discussions

- Basically, robots with closed-loop mechanisms
- Often, contrary to serial robots, actuators are either fixed to the base, or close to it

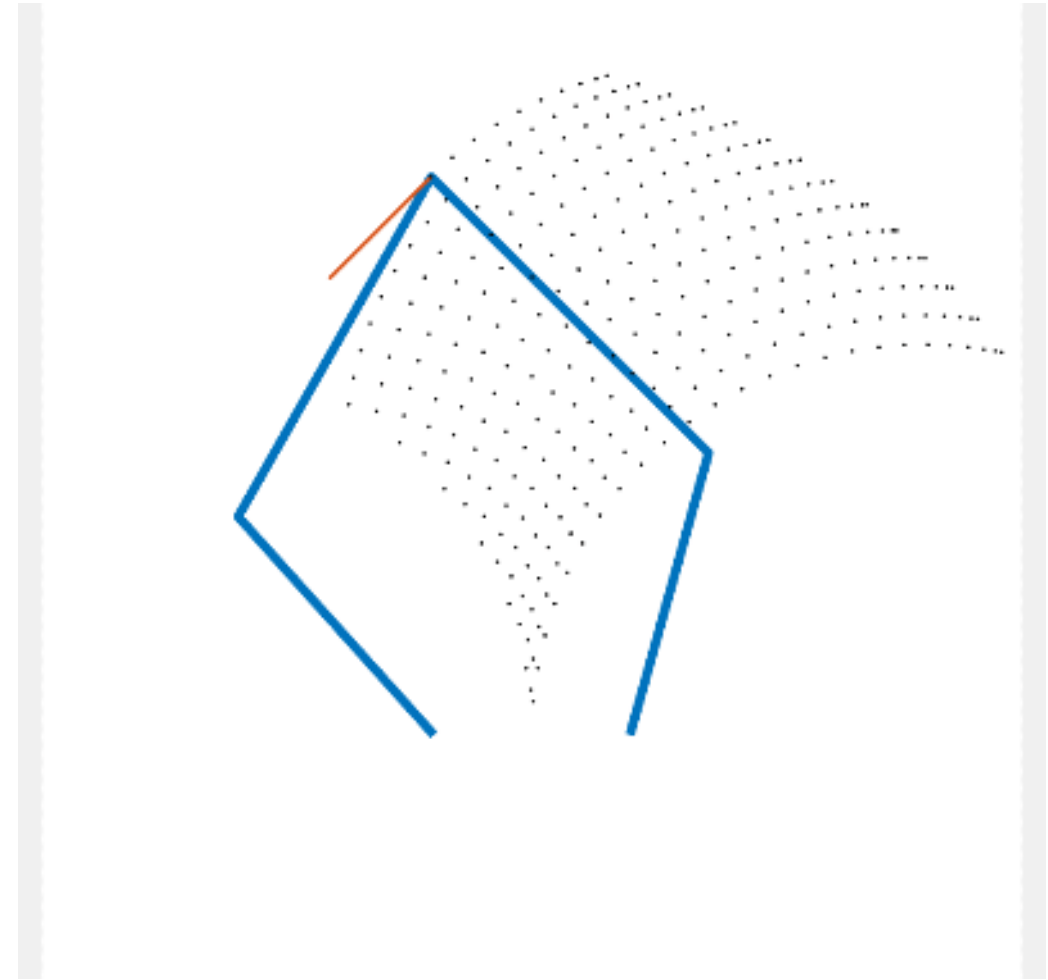


- 3-RRR
- How does it move?
- How many DOFs?
- How to actuate?
- Serial equivalent?



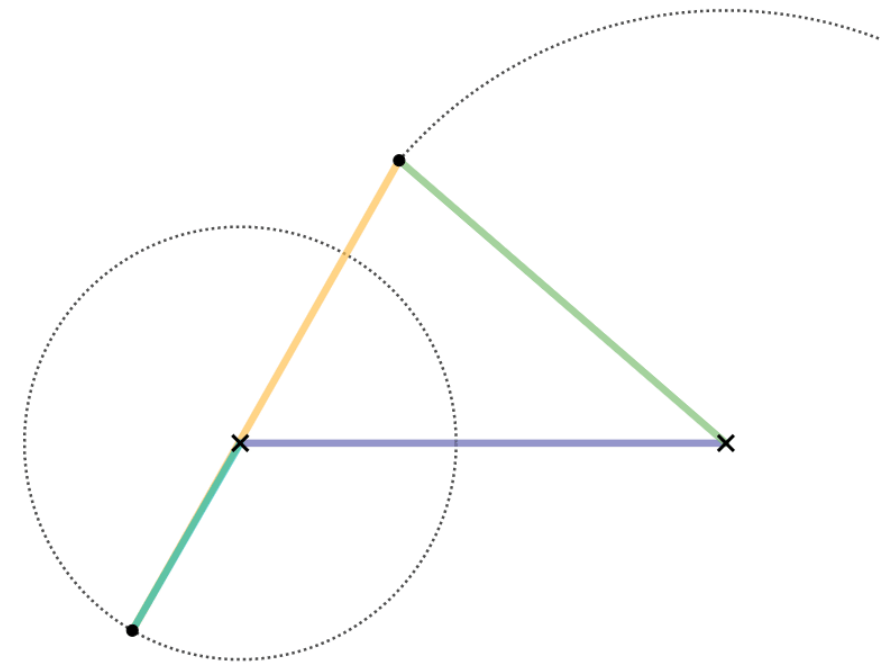
<https://doi.org/10.3390/a11010007>

- 5-bar
- Why 5?
- How does it move?
- How many DOFs?
- How to actuate?
- Serial equivalent?



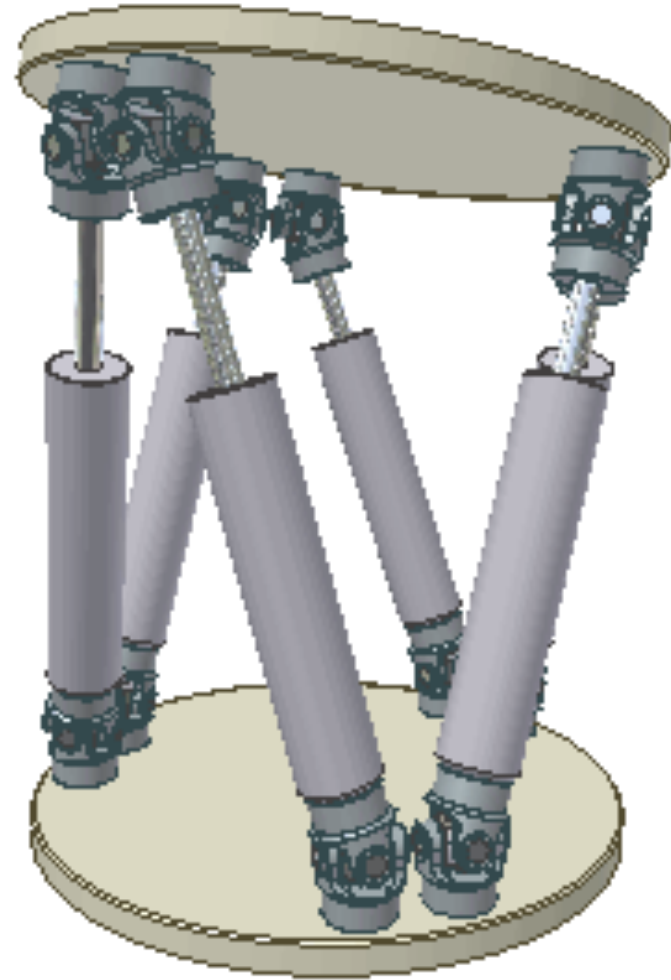
https://en.wikipedia.org/wiki/Five-bar_linkage

- 4-bar
- Is this a robot?
- How does it move?
- How many DOFs?
- How to actuate?
- Much more complex than a serial 1-DOF “system”



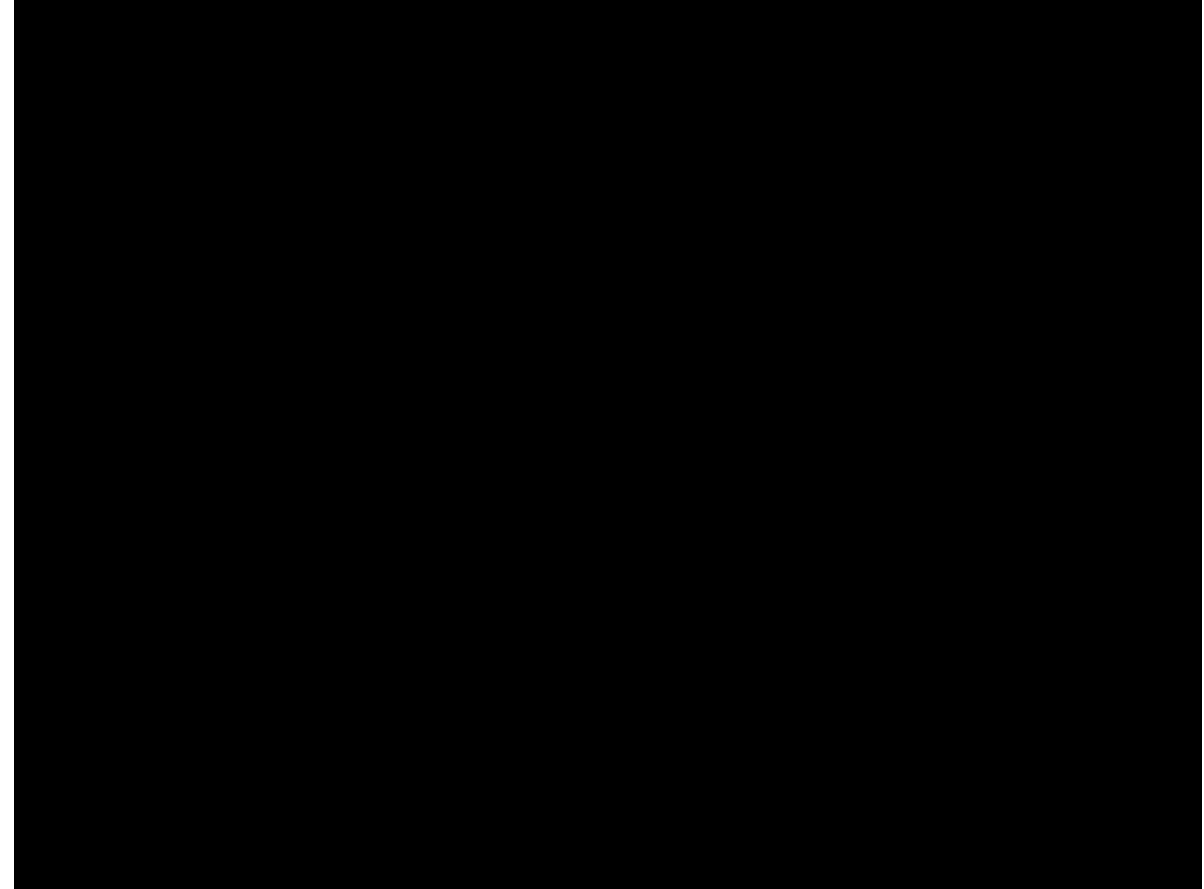
https://en.wikipedia.org/wiki/Four-bar_linkage

- Gough-Stewart platform
- How does it move?
- How many DOFs?
- How to actuate? Why?
- Serial equivalent?

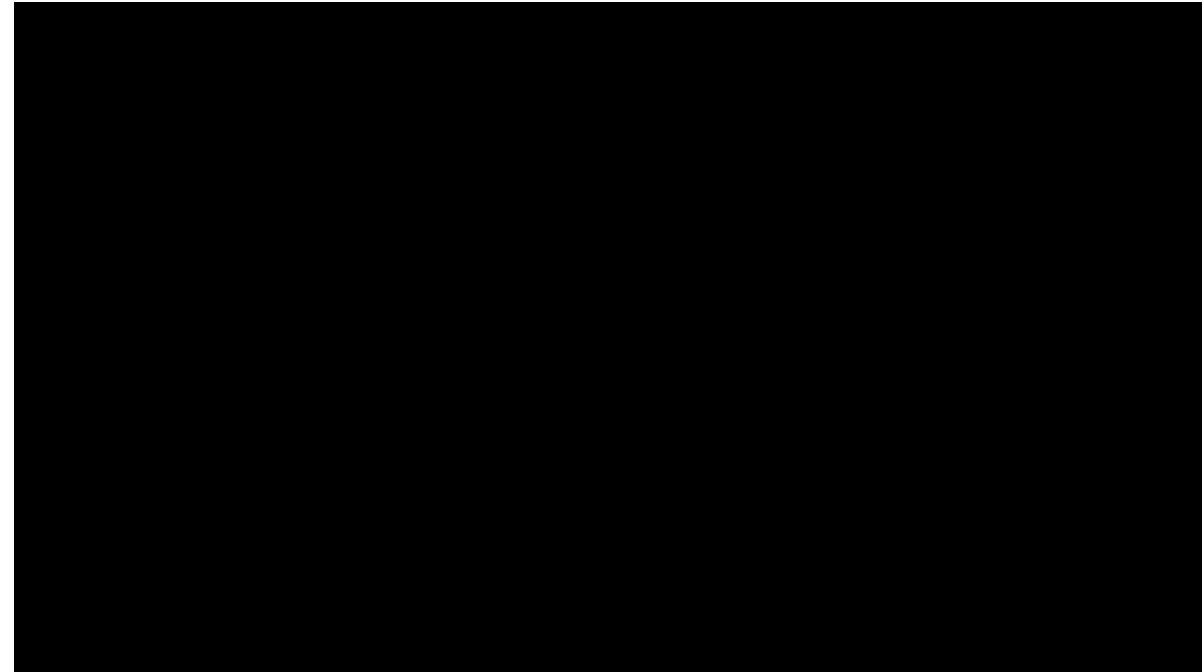


https://en.wikipedia.org/wiki/Stewart_platform

- Hexa robot
- How does it move?
- How many DOFs?
- How to actuate?
- Serial equivalent?



- Parallel Schönflies motion
- How does it move?
- How many DOFs?
- How to actuate?
- Serial equivalent?

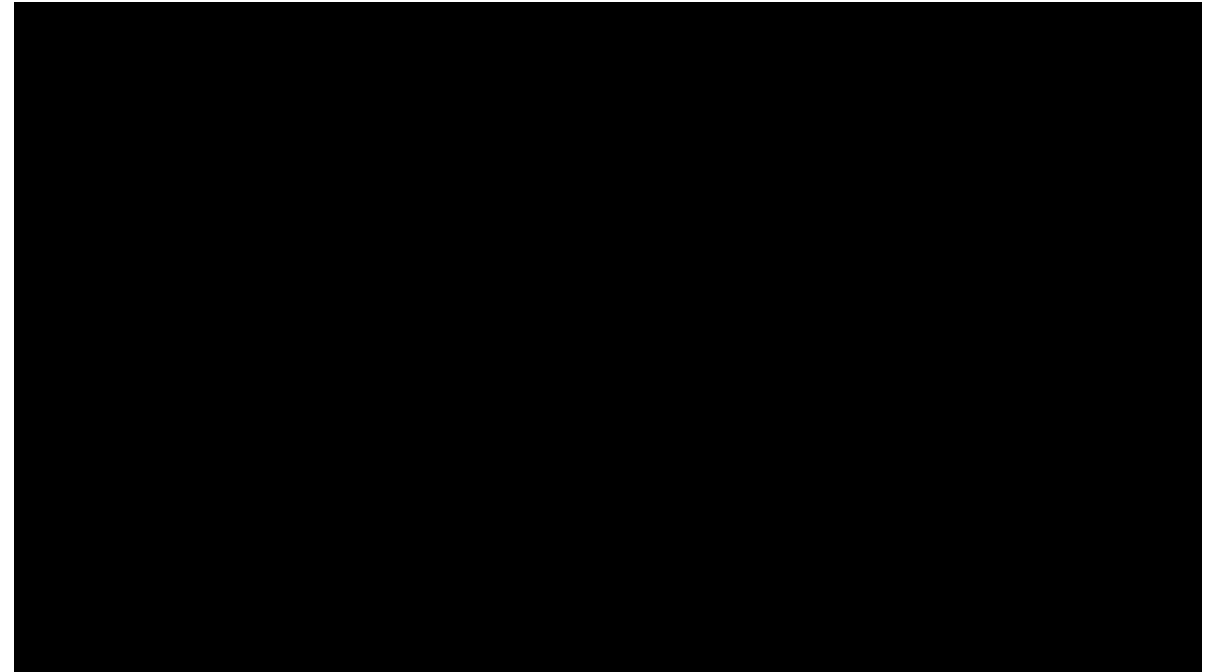


- Delta robot
- How does it move?
- How many DOFs?
- How to actuate?
- Serial equivalent?

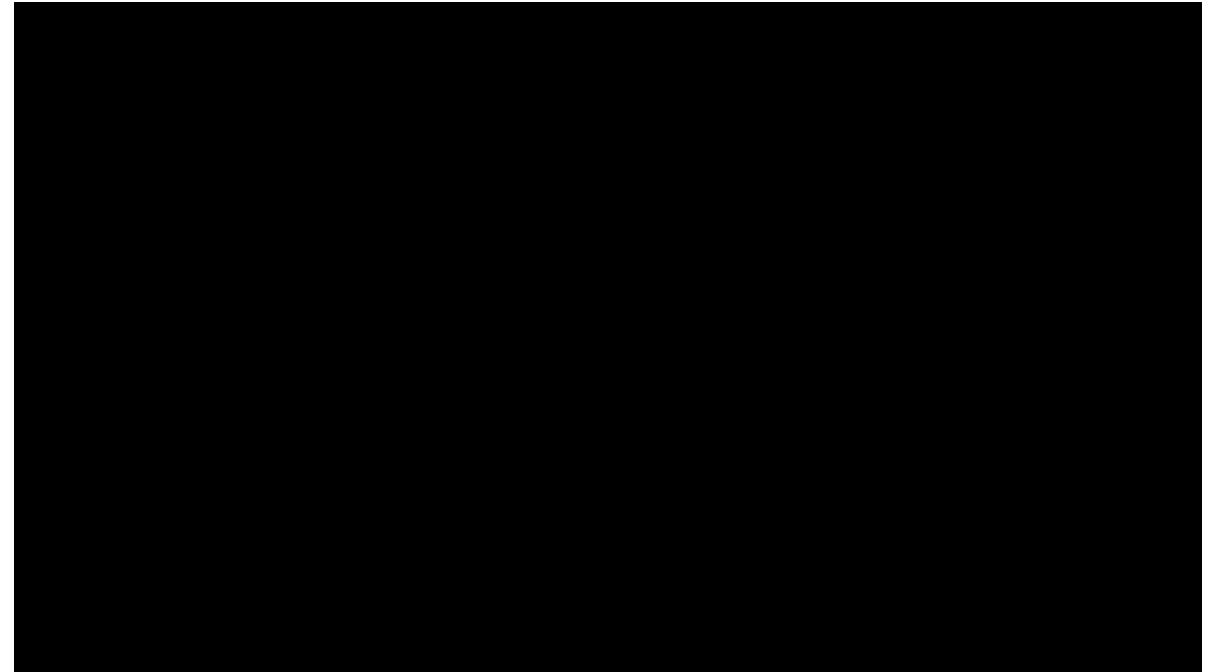


<https://actu.epfl.ch/news/l-epfl-salue-la-memoire-du-pionnier-de-la-robotiqu/>

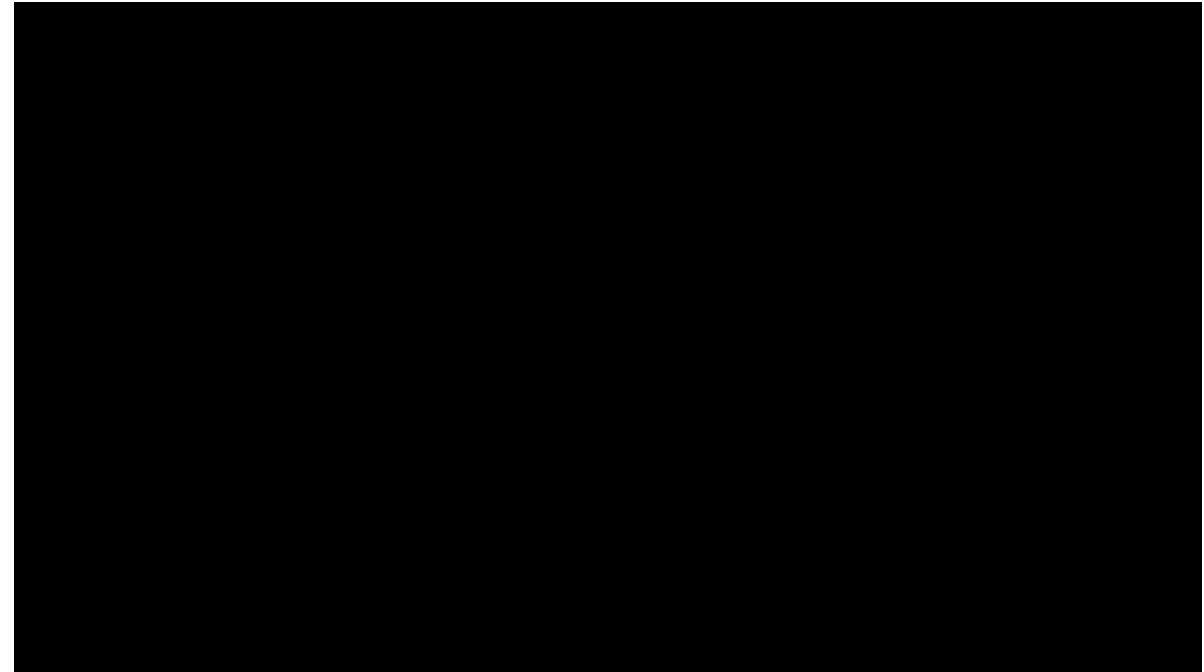
- Kinematically redundant
- How does it move?
- How many DOFs?
- How to actuate?
- Serial equivalent?



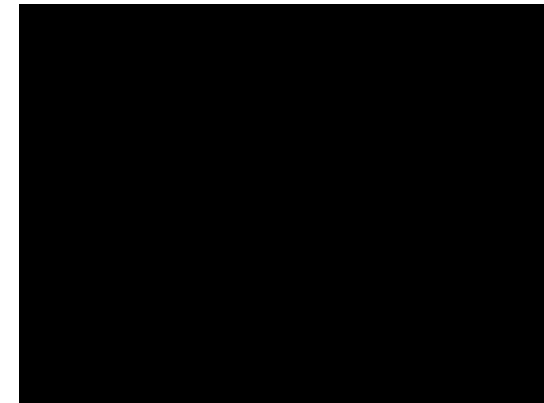
- Actuation redundant
- How does it move?
- How many DOFs?
- How to actuate?
- Serial equivalent?



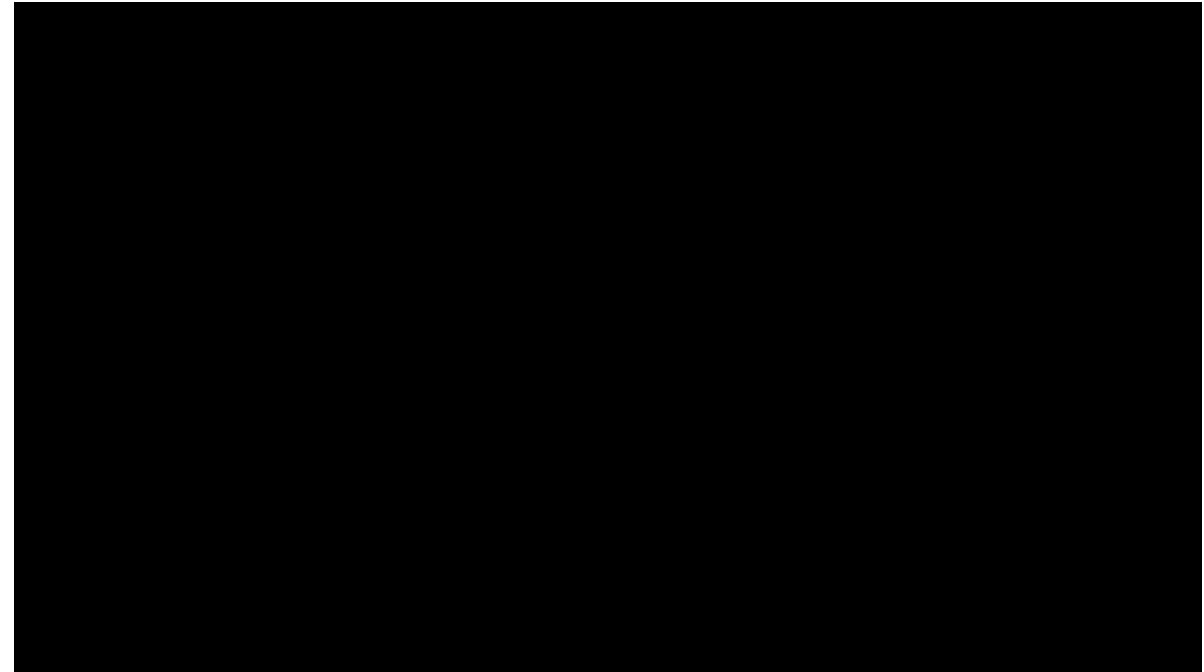
- Cable-driven parallel robot (CDPR)
- How does it move?
- How many DOFs?
- How to actuate?
- Difference w.r.t. previous examples?



- Flying parallel robots
- How does it move?
- How many DOFs?
- How to actuate?



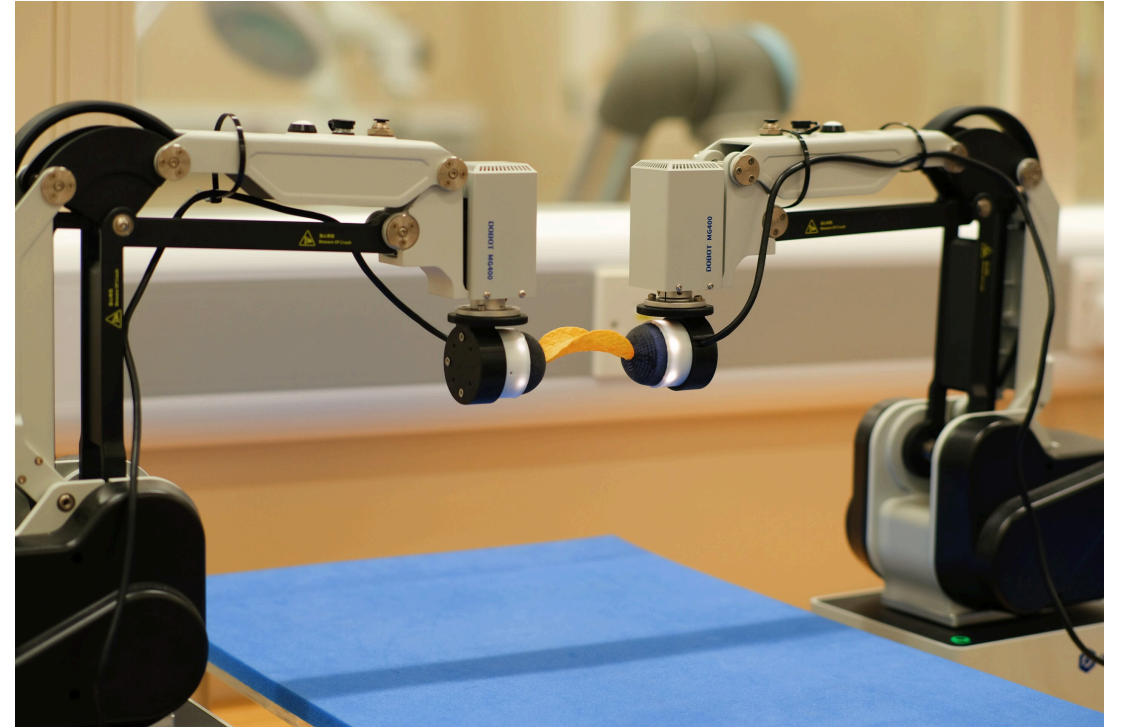
- Continuum parallel robots
- How does it move?
- How many DOFs?
- How to actuate?
- Difference w.r.t. rigid examples?
- How to model?



Other examples?



<https://actu.epfl.ch/news/spot-de-boston-dynamics-chez-robosphere-a-la-chaux/>



<https://www.eurekalert.org/news-releases/999198>

- Parallel robots are, in general, more “complex” than serial ones
 - They share the challenges of serial robots
 - They have their own problems

- So, why do they exist?

1. Actuation

- Actuated at the base with more powerful motors
- Limit case: micro robots cannot carry motors

2. Performance

- Stiffness (lower lever arm)
- Speed (lower inertia)
- Reflected inertia (direct-drive)
- Accuracy (debates)

- Parallel robots can do things that serial robots cannot
- Parallel and serial robots are dual and complementary
- While parallel robots are historically the first ones, serial robots are much more popular. One reason: versatility
 - Parallel robots are designed for a specific task, and perform poorly for most of other tasks
 - Serial robots have suboptimal behavior for everything, but rarely perform so poorly
- Exceptions: Delta and Schönflies motion parallel robots